

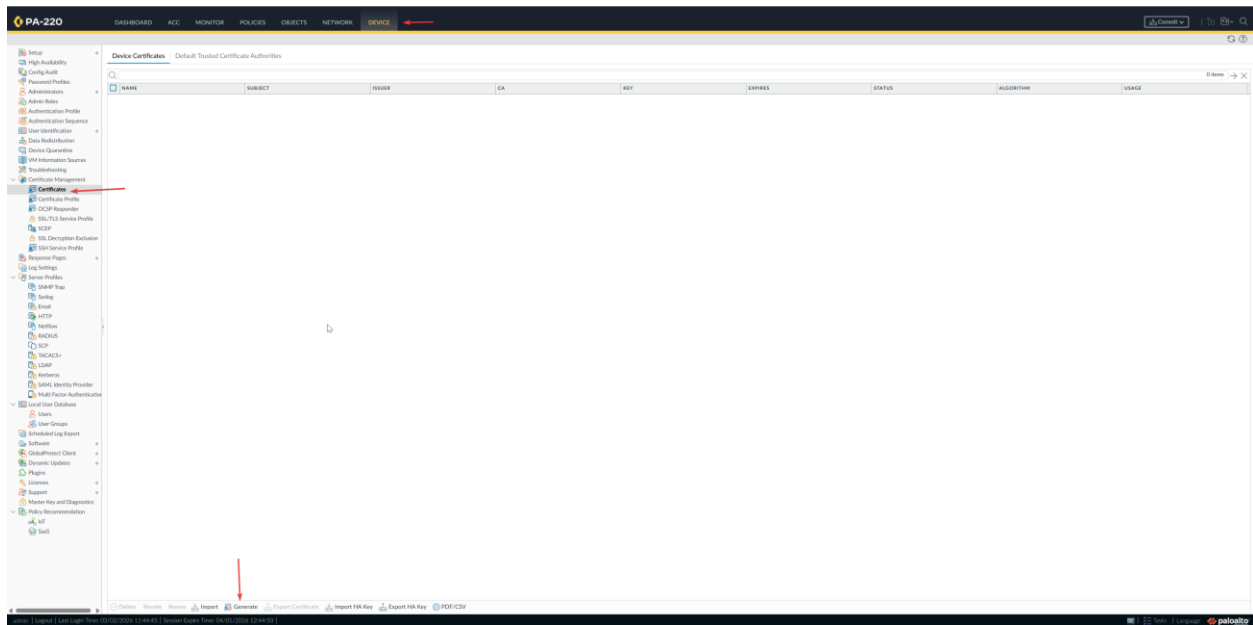
Purpose:

This guide/analysis

Lab Commands

Generate Certificates

In the PA220 web interface, navigate to Device > Certificate Management > Certificates and click Generate.



First, generate a root certificate. Give a Certificate Name and Common Name that make sense for the context, and make sure that the Certificate Authority box is checked.

Under Certificate Attributes, assign appropriate values for the Country, State, Locale, Organization, and Department Fields, and make sure that the IP Address field is set to the outward-facing IP of the firewall.

Generate Certificate

Certificate Type

☒ Local ☐ SCEP

Certificate Name

GlobalProtectStrongerRoot

Common Name

GlobalProtectStrongerRoot

IP or FQDN to appear on the certificate

Signed By

☒ Certificate Authority ☐ Block Private Key Export

OCSP Responder

Cryptographic Settings

Algorithm

RSA

Number of Bits

4096

Digest

sha512

Expiration (days)

365

Certificate Attributes

TYPE	VALUE
☐ Organization = "O" from "Subject" field	Newport High School
☐ Department = "OU" from "Subject" field	Palo Alto Networks Cybersecurity Academy
☒ IP = "IP Address" from Subject Alternative Name (SAN) field	192.168.40.180

Add

Delete

Generate

Cancel

Next, generate your server certificate. The Common Name is the outward-facing IP address of the firewall, and that the certificate is signed by the root certificate you generated earlier.

Generate Certificate

Certificate Type

☒ Local☐ SCEP

Certificate Name

GlobalProtectStrongerServer

Common Name

192.168.40.180

IP or FQDN to appear on the certificate

Signed By

GlobalProtectStrongerRoot

☐ Certificate Authority☐ Block Private Key Export

OCSP Responder

Cryptographic Settings

Algorithm

RSA

Number of Bits

4096

Digest

sha512

Expiration (days)

365

Certificate Attributes

	TYPE	VALUE
☐	Organization = "O" from "Subject" field	Newport High School
☐	Department = "OU" from "Subject" field	Palo Alto Networks Cybersecurity Academy
☒	IP = "IP Address" from Subject Alternative Name (SAN) field	192.168.40.180

+ Add

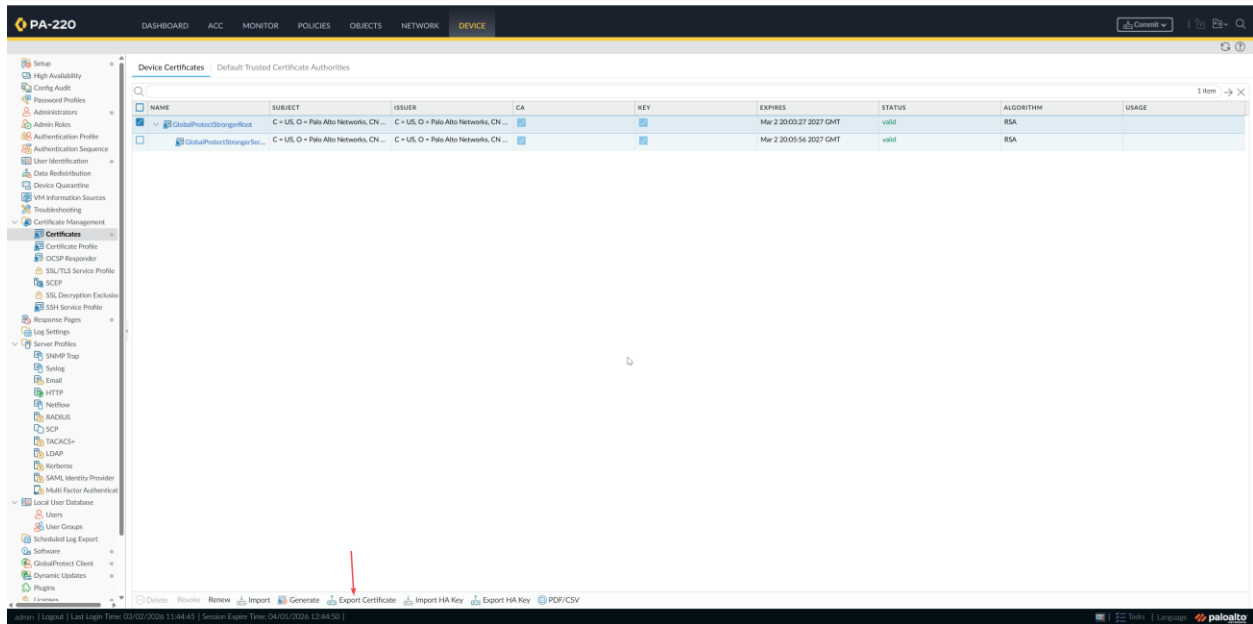
- Delete

Generate

Cancel

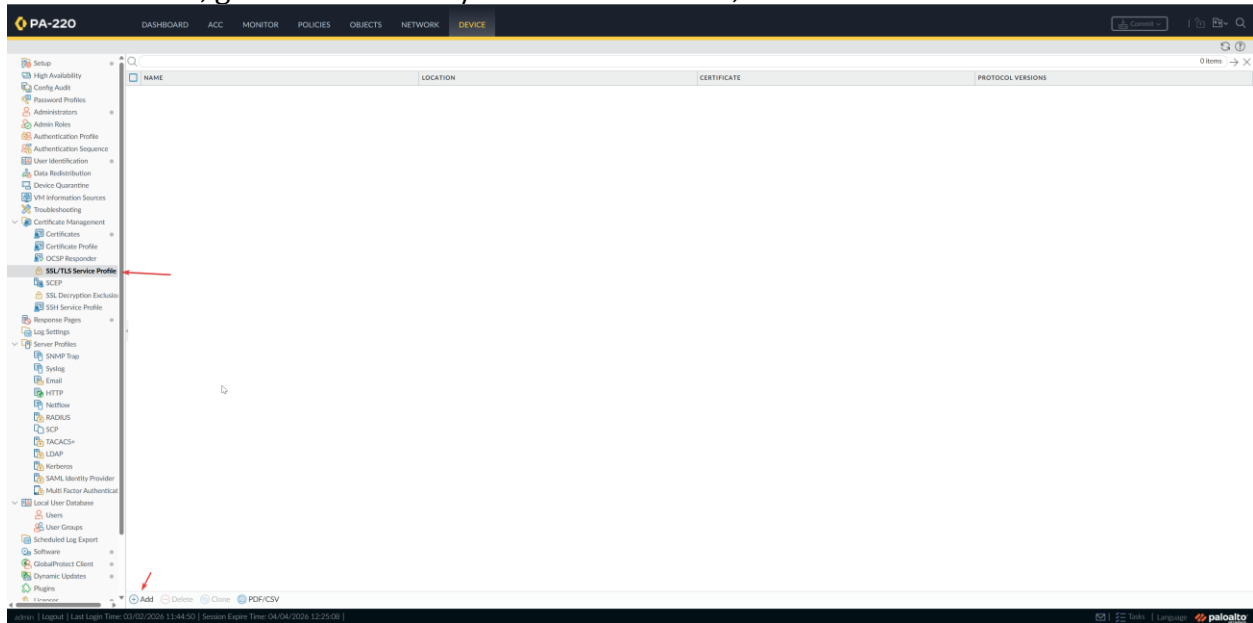
Export your root certificate by selecting it and clicking Export Certificate.

Click “OK” to confirm downloading the certificate. Make sure to note down the file’s download location.



Firewall Configuration

On The PA220, go to Device > SSL/TLS Service Profile, and click “Add”.



Make sure this profile uses the server certificate you set up earlier. \

SSL/TLS Service Profile ?

Name

Certificate

Protocol Settings

Min Version

Max Version

Next, go to “Authentication Profile” (still under Device) and click “Add”.

The screenshot shows the Palo Alto Networks PA-220 configuration interface. The left sidebar contains a tree view of configuration categories. Under the 'Device' category, the 'Authentication Profile' option is highlighted with a red arrow. The main pane displays a table with columns: NAME, LOCATION, FAILED ATTEMPTS (F), LOCKOUT TIME (MIN), ALLOW LIST, AUTHENTICATION, SERVER PROFILE, AUTHENTICATION FACTORS, OTHERS, and LOCKED USERS. The table is currently empty. At the bottom of the sidebar, there are buttons for 'Add', 'Delete', 'Clone', and 'PDF/CSV'. A red arrow points to the 'Add' button. The top of the interface shows the 'PA-220' header and various navigation tabs like DASHBOARD, ACC, MONITOR, POLICIES, OBJECTS, NETWORK, and DEVICE. The bottom status bar shows the user is logged in as 'admin' and the system is running version 6.0.10.

Give this profile an appropriate name and set the database type to “Local Database”.

Authentication Profile ⓘ

Name

Authentication | Factors | **Advanced**

→ Type

User Domain

Username Modifier

Single Sign On

Kerberos Realm

Kerberos Keytab [X Import](#)

Under “Advanced”, click “Add” and type “all”.

Authentication Profile ⓘ

Name

Authentication | Factors | **Advanced**

Allow List

☐	ALLOW LIST ^
☒	<u>all</u>

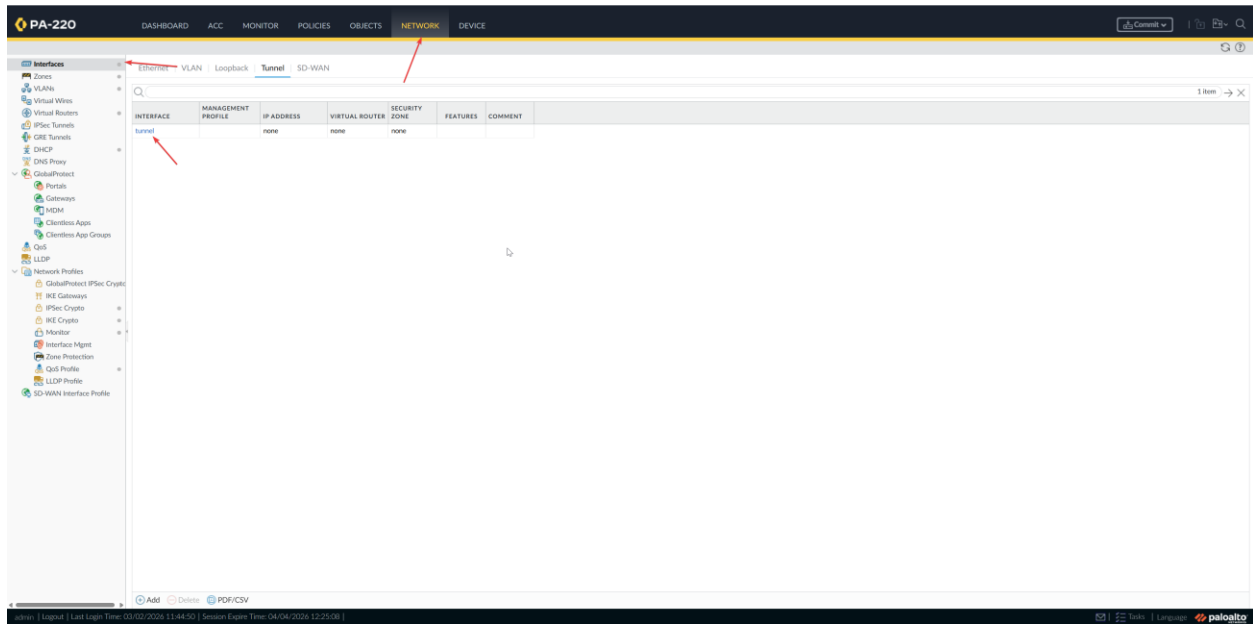
←

Account Lockout

Failed Attempts

Lockout Time (min)

Go to Network > Interfaces > Tunnel and click on the “tunnel” interface.



Under “Config”, configure the same virtual router and Layer 3 security zone used on your inside network.

Tunnel Interface

Interface Name:

Comment:

Netflow Profile:

Config | IPv4 | IPv6 | Advanced

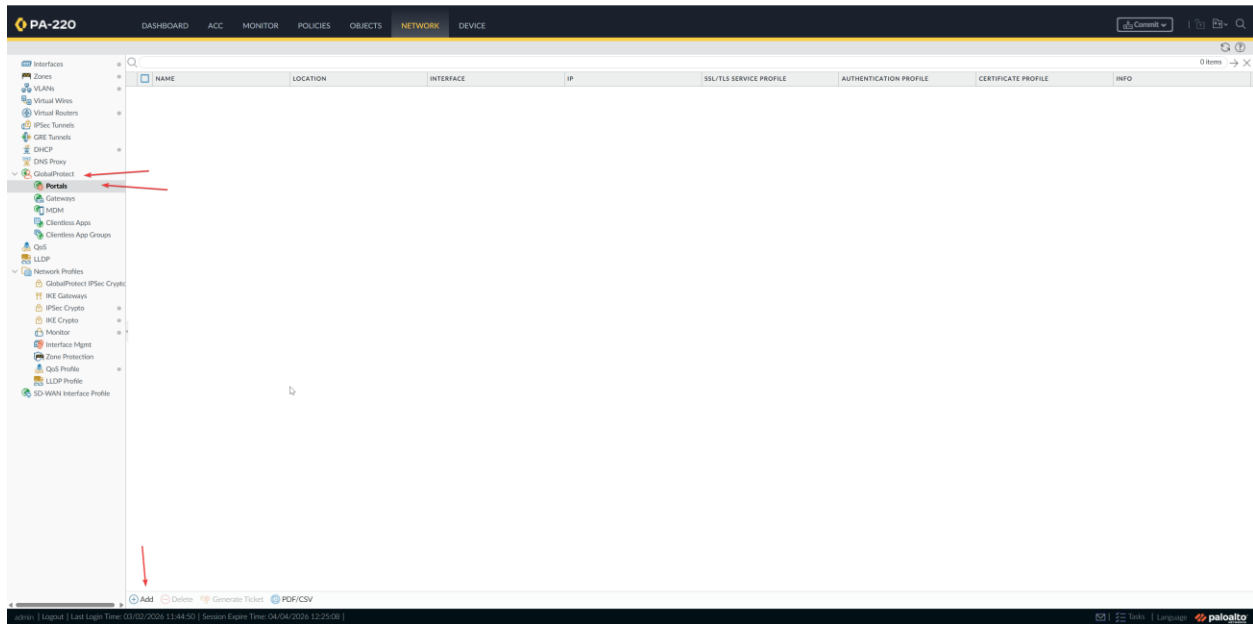
Assign Interface To

Virtual Router:

Security Zone:

OK Cancel

Go to Network > GlobalProtect > Portals and click “Add”.



Specify the outward-facing ethernet interface and set the address type to “IPv4 only”.

External Gateway ?

Name External

Address ☐ FQDN ☒ IP

IPv4 192.168.40.54

IPv6

1 item → ×

☐	SOURCE REGION	PRIORITY
☒	<u>Any</u>	Highest

☒ Add ☐ Delete

☐ Manual (The user can manually select this gateway)

OK Cancel

Under the “Authentication” tab, specify the SSL/TLS service profile you created earlier and click “Add”.²¹

GlobalProtect Portal Configuration

General

Authentication

Portal Data Collection

Agent

Clientless VPN

Satellite

Server Authentication

SSL/TLS Service Profile GlobalProtect

Client Authentication

	NAME	OS	AUTHENTIC...	AUTO RETRIEVE PASSCODE	USERNAME LABEL	PASSWORD LABEL	AUTHENTI... MESSAGE	ALLOW AUTHENTI... WITH USER CREDENTI... OR CLIENT CERTIFICA...
☐								

+ Add

- Delete

🔄 Clone

↑ Move Up

↓ Move Down

Certificate Profile None

OK

Cancel

Configure an appropriate tunnel name, and set the authentication profile to the profile created earlier. Configure an appropriate authentication message, and allow authentication with user credentials OR a client certificate.

Client Authentication

Name tunnel-1

OS Any

Authentication Profile Local Auth

☐ Automatically retrieve passcode from SoftToken application

GlobalProtect App Login Screen

Username Label Username

Password Label Password

Authentication Message Please Authenticate

Authentication message can be up to 256 characters.

Allow Authentication with User Credentials OR Client Certificate Yes (User Credentials OR Client Certificate Required)

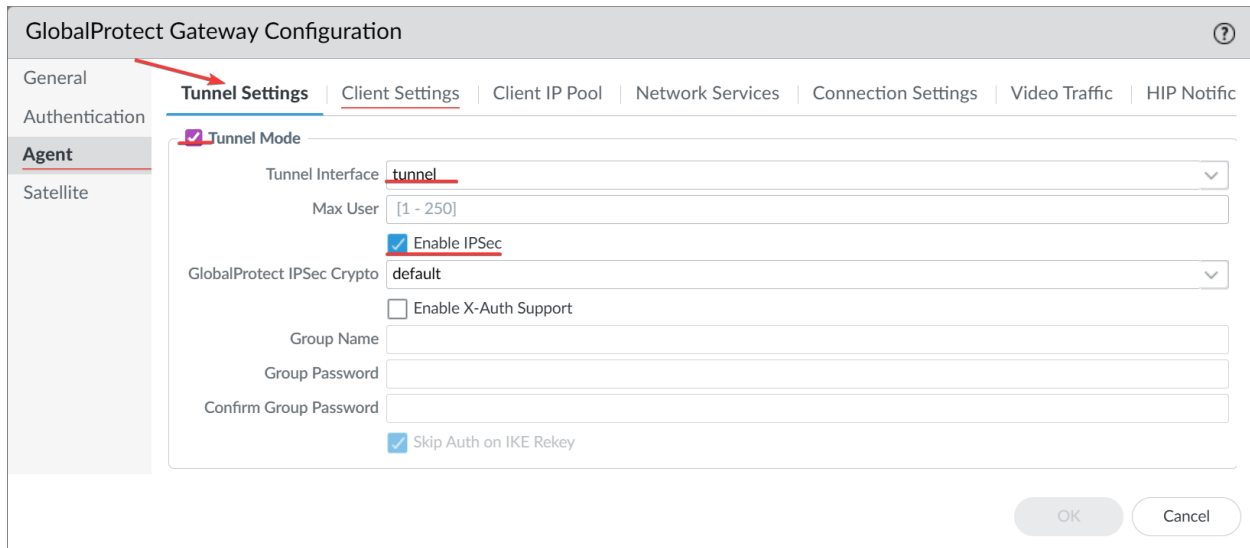
To enforce client certificate authentication, you must also select the certificate profile in the Client Authentication configuration.

OK

Cancel

Under Agent > Tunnel Settings, set the tunnel interface to the interface created earlier,

and make sure to enable IPsec.



The screenshot shows the 'GlobalProtect Gateway Configuration' window with the 'Tunnel Settings' tab selected. A red arrow points to the 'Tunnel Settings' tab. The 'Agent' section is expanded in the left sidebar. The 'Tunnel Mode' checkbox is checked. The 'Tunnel Interface' dropdown is set to 'tunnel'. The 'Max User' field is '[1 - 250]'. The 'Enable IPsec' checkbox is checked and highlighted with a red underline. The 'GlobalProtect IPsec Crypto' dropdown is set to 'default'. The 'Enable X-Auth Support' checkbox is unchecked. The 'Group Name', 'Group Password', and 'Confirm Group Password' fields are empty. The 'Skip Auth on IKE Rekey' checkbox is checked. The 'OK' and 'Cancel' buttons are at the bottom right.

GlobalProtect Gateway Configuration

General | **Tunnel Settings** | Client Settings | Client IP Pool | Network Services | Connection Settings | Video Traffic | HIP Notific

Authentication

Agent

Satellite

☒ Tunnel Mode

Tunnel Interface: tunnel

Max User: [1 - 250]

☒ Enable IPsec

GlobalProtect IPsec Crypto: default

☐ Enable X-Auth Support

Group Name:

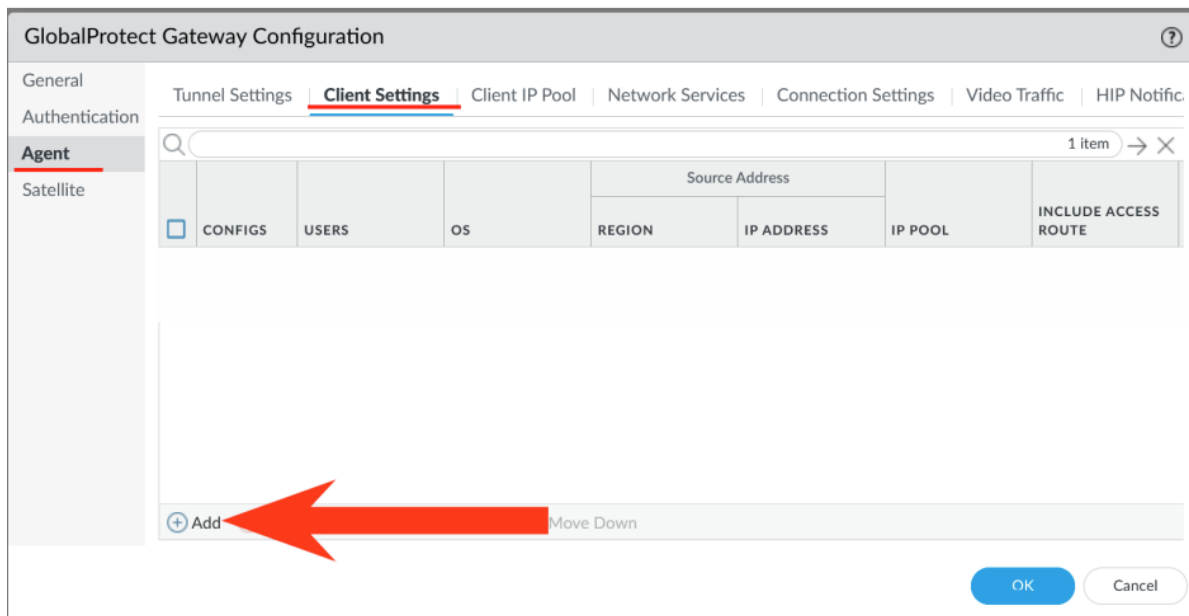
Group Password:

Confirm Group Password:

☒ Skip Auth on IKE Rekey

OK Cancel

Under “Client Settings”, click “Add”.



The screenshot shows the 'GlobalProtect Gateway Configuration' window with the 'Client Settings' tab selected. The 'Agent' section is expanded in the left sidebar. The 'Client Settings' tab shows a table with columns: CONFIGS, USERS, OS, REGION, IP ADDRESS, IP POOL, and INCLUDE ACCESS ROUTE. The table is currently empty. A red arrow points to the 'Add' button at the bottom left of the table. The 'OK' and 'Cancel' buttons are at the bottom right.

GlobalProtect Gateway Configuration

General | Tunnel Settings | **Client Settings** | Client IP Pool | Network Services | Connection Settings | Video Traffic | HIP Notific

Authentication

Agent

Satellite

1 item → ×

	CONFIGS	USERS	OS	Source Address		IP POOL	INCLUDE ACCESS ROUTE
				REGION	IP ADDRESS		

+ Add Move Down

OK Cancel

Give the configuration an appropriate name, and set the config selection criteria to “any”.

Configs ?

Config Selection Criteria | Authentication Override | IP Pools | Split Tunnel | Network Services

Name: Auth-override

Config Selection Criteria

any	☒ Any
☐ SOURCE USER ^	☐ OS ^
+ Add - Delete	+ Add - Delete

Source Address

☐ REGION ^	☐ IP ADDRESS ^
+ Add - Delete	+ Add - Delete

The configuration must match User and OS and either Region or IP Address if specified.

OK **Cancel**

Under “Authentication Override”, set the certificate to decrypt/encrypt the cookie to the root certificate created earlier.

Configs ?

Config Selection Criteria | **Authentication Override** | IP Pools | Split Tunnel | Network Services

☒ Generate cookie for authentication override

☒ Accept cookie for authentication override

Cookie Lifetime: Hours

Certificate to Encrypt/Decrypt Cookie: GlobalProtectStrongerRoot

OK **Cancel**

Under “IP Pools”, click “Add” and specify a valid range of IP addresses. These addresses will be dynamically allocated to clients by the GlobalProtect gateway.

⑦

☐ Retrieve Framed-IP-Address attribute from authentication server

These IPs will be added to the firewall's routing table

Cancel



Configure an appropriate username and password

Local User ?

Name

Mode ☒ Password ☐ Password Hash

Password

Confirm Password

☒ Enable

Under GlobalProtect Client, download the latest client version.

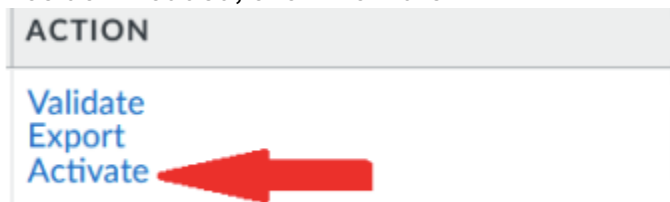
PA-220 DASHBOARD ACC MONITOR POLICIES OBJECTS NETWORK **DEVICE**

	VERSION	SIZE	RELEASE DATE	AVAILABLE	CURRENTLY INSTALLED	ACTION	
Admin Roles							
Authentication Profile							
Authentication Sequence							
User Identification	6.3.3-0876	274 MB	2026/02/03 09:37:24			Download	Release Notes
Data Redistribution	6.3.3-0842	251 MB	2025/12/17 02:07:54			Download	Release Notes
Device Quarantine	6.3.3-0828	251 MB	2025/11/20 03:25:29			Download	Release Notes
VM Information Sources	6.3.3-0711	251 MB	2025/09/22 14:02:59			Download	Release Notes
Trustedhosting	6.3.3-0676	251 MB	2025/07/28 04:52:56			Download	Release Notes
Certificate Management	6.3.3	251 MB	2025/04/30 07:44:28			Download	Release Notes
Certificates	6.3.3-0450	251 MB	2025/04/12 03:21:09			Download	Release Notes
Certificate Profile	6.3.2	235 MB	2024/12/23 08:32:25			Download	Release Notes
OCSP Responder	6.3.1-0803	224 MB	2024/10/14 12:11:28			Download	Release Notes
SSL/TLS Service Profile	6.3.1	224 MB	2024/09/11 13:20:16			Download	Release Notes
SCP	6.3.0	224 MB	2024/06/13 10:50:28			Download	Release Notes
SSL Decryption Exclusion	6.2.8-0471	249 MB	2024/02/03 09:09:23			Download	Release Notes
SSH Service Profile	6.2.8-0416	248 MB	2025/11/18 04:32:50			Download	Release Notes
Response Pages	6.2.8-0317	247 MB	2025/09/09 09:00:09			Download	Release Notes
Log Settings	6.2.8-0263	247 MB	2025/07/17 07:35:16			Download	Release Notes
Server Profiles	6.2.8-0243	247 MB	2025/06/23 06:47:15			Download	Release Notes
SNMP Trap	6.2.8	232 MB	2025/04/03 10:48:56			Download	Release Notes
Spring	6.2.8-0201	248 MB	2025/12/01 09:54:15			Download	Release Notes
Email	6.2.8-0223	246 MB	2025/05/15 08:49:36			Download	Release Notes
HTTP	6.2.7	230 MB	2025/01/28 11:40:30			Download	Release Notes
Netflow	6.2.6-0857	230 MB	2025/01/16 07:54:09			Download	Release Notes
RADIUS	6.2.6	229 MB	2024/11/12 16:15:08			Download	Release Notes
SCP	6.2.5-0788	229 MB	2024/10/14 04:44:35			Download	Release Notes
TACACS+	6.2.4	222 MB	2024/07/15 11:35:34			Download	Release Notes
LDAP	6.2.3-0287	230 MB	2024/04/04 10:52:11			Download	Release Notes
Yerkes	6.2.3	220 MB	2024/04/10 05:58:14			Download	Release Notes
SAML Identity Provider	6.2.2	209 MB	2023/11/22 05:58:19			Download	Release Notes
Multi Factor Authentication	6.2.1	209 MB	2023/10/09 07:22:05			Download	Release Notes
Local User Database	6.2.0	257 MB	2023/05/22 07:45:59			Download	Release Notes
Users	6.1.5	219 MB	2024/06/20 13:11:50			Download	Release Notes
User Groups	6.1.4-0720	211 MB	2024/10/23 03:45:52			Download	Release Notes
Scheduled Log Export	6.1.4	211 MB	2024/01/29 02:43:59			Download	Release Notes
Software	6.1.3	292 MB	2023/11/21 08:28:57			Download	Release Notes
Dynamic Updates	6.1.2	290 MB	2023/08/03 07:15:29			Download	Release Notes
Plugins	6.1.1	250 MB	2023/03/14 07:32:48			Download	Release Notes
Licenses	6.1.0	124 MB	2022/09/01 14:06:19			Download	Release Notes
Support	6.0.12	232 MB	2025/09/02 10:18:58			Download	Release Notes
Master Key and Diagnostics							
Policy Recommendation							
Int							
Int							

48 Items

admin | Logout | Last Login Time: 03/02/2025 11:44:50 | Session Expires Time: 04/04/2025 12:25:08

Once downloaded, click “Activate”.



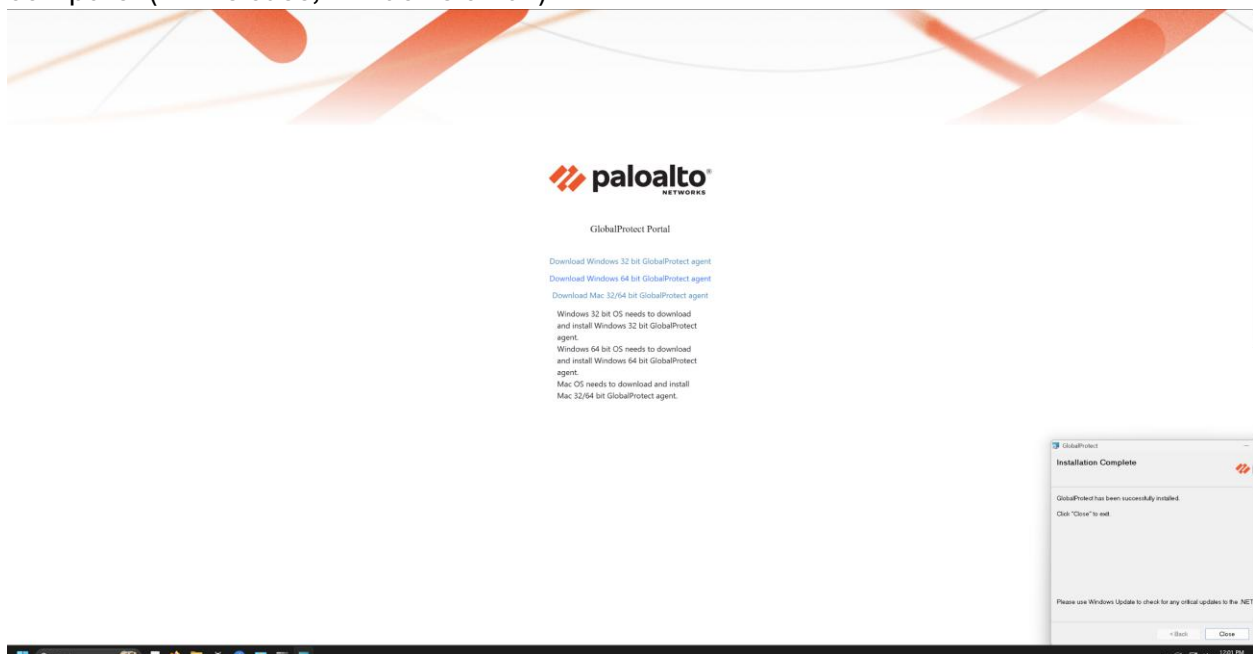
Connecting to the VPN

Switch to the outside computer. From a web browser, navigate to the outward-facing IP of the firewall.

Sign In to your Test-User

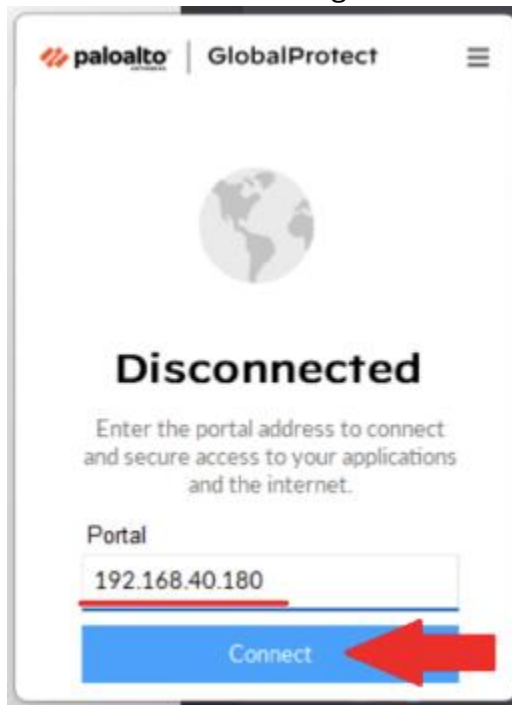


Download the appropriate GlobalProtect agent for the OS/architecture of the outside computer (in this case, Windows 64 bit).



Click "Next" through the installation process.
Open the GlobalProtect client from the system tray (it may be in the overflow section).

Enter the outward-facing IP of the firewall and click “Connect”



If connected, you will see this success message

